A Non-Field Stable Serial Knockout Competition on Sixteen Players

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Abstract

Lambers, Pendavingh, and Spieksma constructed stable serial knockout competitions on 2^k players using the finite field $\mathrm{GF}(2^k)$, and asked whether, for $2^k > 8$, every stable serial knockout competition is equivalent to their construction. This note gives an explicit negative answer already for $n = 16$. The construction is a family of 15 knockout brackets on the additive group $\mathbb{F}_2^4$. For each nonzero translation difference δ, the number of brackets in which a pair at difference δ can meet in rounds $1, 2, 3, 4$ is exactly $(1, 2, 4, 8)$. The construction is not equivalent to the finite-field construction because two of its brackets have the same round-two partition, whereas the finite-field construction has fifteen distinct round-two partitions.

1 Definitions

Let P be a set of $n = 2^k$ players. A balanced knockout tournament, or bracket, on P determines a map

$$v_T(x, y) \in \{1, \dots, k\} \qquad (x \neq y),$$

where $v_T(x, y)$ is the round in which x and y can meet, assuming suitable outcomes of earlier matches. A serial knockout competition (SKC) is a set $\mathcal{T}$ of $n - 1$ such brackets on the same player set. It is *stable* if, for each round i, the number

$$\#\{T \in \mathcal{T} : v_T(x, y) = i\}$$

is independent of the unordered pair $\{x, y\}$. When $|\mathcal{T}| = n - 1$, the stable round counts must be

$$1, 2, 4, \dots, 2^{k-1},$$

because, in one bracket, each player can potentially meet 2^{i-1} opponents in round i.

Two SKCs are equivalent if one can be obtained from the other by relabeling the players. The finite-field construction of Lambers–Pendavingh–Spieksma gives a stable SKC for every $n = 2^k$ [1]. The question addressed here is whether that construction is unique up to relabeling for $n > 8$. The following theorem gives a counterexample for $n = 16$.

Theorem 1. *There exists a stable SKC on* 16 *players that is not equivalent to the Lambers–Pendavingh–Spieksma finite-field construction.*

2 The construction

Let

$$V = \mathbb{F}_2^4.$$

We label the elements of V by hexadecimal symbols

$$0, 1, 2, \ldots, 9, a, b, c, d, e, f,$$

using bitwise xor as addition. For example, $2 + 5 = 7$ and $a + 6 = c$. A string such as `123` denotes the set $\{1, 2, 3\}$, and a string such as `12389ab` denotes $\{1, 2, 3, 8, 9, a, b\}$.

For each nonzero $p \in V$, define a bracket T_p by a flag of additive subspaces

$$S_p \subset U_p \subset W_p \subset V, \qquad \dim S_p = 1, \quad \dim U_p = 2, \quad \dim W_p = 3.$$

The first-round pairs of T_p are the cosets of S_p, the round-two blocks of four players are the cosets of U_p, and the semifinal halves of eight players are the cosets of W_p.

We always set

$$S_p = \{0, p\}.$$

The subspaces U_p and W_p are specified in Table 1: if the row gives L_p and H_p, then

$$U_p = \{0\} \cup L_p, \qquad W_p = \{0\} \cup H_p.$$

The last column gives a normal vector ν_p for the hyperplane W_p, with respect to the usual dot product on $\mathbb{F}_2^4$:

$$W_p = \{x \in V : \nu_p \cdot x = 0\}.$$

p	L_p	H_p	ν_p
1	123	12389ab	4
2	123	1234567	8
3	3cf	123cdef	c
4	4ae	2468ace	1
5	257	2578adf	5
6	69f	16789ef	6
7	7ad	3479ade	b
8	68e	3568bde	7
9	59c	3569acf	f
a	1ab	145abef	a
b	6bd	167abcd	e
c	48c	3478bcf	3
d	58d	14589cd	2
e	79e	2579bce	d
f	4bf	2469bdf	9

Table 1: The flags defining the fifteen brackets. Each row defines $S_p = \{0, p\}$, $U_p = \{0\} \cup L_p$, and $W_p = \{0\} \cup H_p$.

Lemma 1. *Table 1 defines fifteen valid nested flags $S_p \subset U_p \subset W_p$ with dimensions $1, 2, 3$.*

Proof. For each row, $p \in L_p$ and the three entries of L_p sum to zero in $\mathbb{F}_2^4$. Thus $\{0\} \cup L_p$ is a two-dimensional subspace containing S_p. The last column certifies that $\{0\} \cup H_p$ is the hyperplane perpendicular to ν_p, hence is a three-dimensional subspace. Finally, the inclusion $L_p \subset H_p$ is visible row-by-row in Table 1. Therefore $S_p \subset U_p \subset W_p$ is a flag of the desired dimensions. $\square$

Define

$$\mathcal{T} = \{T_p : p \in V \setminus \{0\}\}.$$

This is a set of fifteen brackets. The brackets are distinct because the first-round subspace S_p, namely the first-round pair containing 0, determines p.

3 Stability

The construction is translation invariant in the following useful sense. If $x, y \in V$ are distinct, put

$$\delta = x + y \neq 0.$$

In the bracket T_p, the players x and y are in the same first-round pair if and only if they are in the same coset of S_p, equivalently $\delta \in S_p$. Similarly, they lie in the same round-two block if and only if $\delta \in U_p$, and they lie in the same semifinal half if and only if $\delta \in W_p$. Therefore

$$\begin{aligned} v_{T_p}(x, y) = 1 &\iff \delta \in S_p \setminus \{0\}, \\ v_{T_p}(x, y) \leq 2 &\iff \delta \in U_p \setminus \{0\}, \\ v_{T_p}(x, y) \leq 3 &\iff \delta \in W_p \setminus \{0\}. \end{aligned}$$

We now count these incidences for a fixed nonzero δ.

First, $\delta \in S_p$ for exactly one value of p, namely $p = \delta$.

Second, $\delta \in U_p$ exactly when $\delta \in L_p$. The following incidence table is a direct certificate that each nonzero vector of V appears in exactly three of the sets L_p:

δ	$\{p : \delta \in L_p\}$	δ	$\{p : \delta \in L_p\}$	δ	$\{p : \delta \in L_p\}$
1	$\{1, 2, \mathtt{a}\}$	2	$\{1, 2, 5\}$	3	$\{1, 2, 3\}$
4	$\{4, \mathtt{c}, \mathtt{f}\}$	5	$\{5, 9, \mathtt{d}\}$	6	$\{6, 8, \mathtt{b}\}$
7	$\{5, 7, \mathtt{e}\}$	8	$\{8, \mathtt{c}, \mathtt{d}\}$	9	$\{6, 9, \mathtt{e}\}$
a	$\{4, 7, \mathtt{a}\}$	b	$\{\mathtt{a}, \mathtt{b}, \mathtt{f}\}$	c	$\{3, 9, \mathtt{c}\}$
d	$\{7, \mathtt{b}, \mathtt{d}\}$	e	$\{4, 8, \mathtt{e}\}$	f	$\{3, 6, \mathtt{f}\}$.

Hence δ lies in exactly three of the U_p's.

Third, $\delta \in W_p$ if and only if $\nu_p \cdot \delta = 0$. The normal vectors ν_p in Table 1 are precisely the fifteen nonzero vectors of V. For fixed nonzero δ, the orthogonal hyperplane

$$\delta^\perp = \{\nu \in V : \nu \cdot \delta = 0\}$$

has eight elements, of which seven are nonzero. Thus δ lies in exactly seven of the W_p's.

It follows that, for every pair $x \neq y$, the number of brackets in which the pair can meet in round 1 is

$$1,$$

the number in which it can meet in round 2 is

$$3 - 1 = 2,$$

the number in which it can meet in round 3 is

$$7 - 3 = 4,$$

and the number in which it can meet in round 4 is

$$15 - 7 = 8.$$

These are exactly the required stable counts for a 16-player SKC. Hence $\mathcal{T}$ is stable.

4 Inequivalence to the finite-field construction

For a bracket T on sixteen players, let $\Pi_2(T)$ denote its partition into four round-two blocks of size 4. The multiset

$$\{\Pi_2(T) : T \in \mathcal{T}\}$$

is an invariant of an SKC up to relabeling: a relabeling merely sends each partition to its relabeled image, and therefore preserves whether two brackets have the same round-two partition.

In the SKC constructed above, the first two rows of Table 1 have the same line,

$$L_1 = L_2 = \texttt{123}.$$

Consequently

$$U_1 = U_2 = \{0, 1, 2, 3\},$$

and T_1 and T_2 have the same round-two partition. Thus the multiset of round-two partitions in this SKC has a repeated element.

We next recall the relevant feature of the finite-field construction. Let $F = \mathrm{GF}(16)$, and let $\alpha \in F$ have degree 4 over $\mathbb{F}_2$. In the Lambers–Pendavingh–Spieksma construction, the round-two partitions are the additive coset partitions of the 15 two-dimensional subspaces

$$zD \qquad (z \in F^*), \qquad D = \langle 1, \alpha\rangle_{\mathbb{F}_2}.$$

These 15 subspaces are all distinct. Indeed, suppose $zD = z'D$ for nonzero z, z'. With $c = z^{-1}z'$, this says $cD = D$. Since $1 \in D$, we have $c \in D$. If $c \neq 1$, then $c \in D \setminus \{0, 1\}$. The equality $cD = D$ implies $c^2 \in D$, so c satisfies a polynomial of degree at most 2 over $\mathbb{F}_2$. But $D \setminus \{0, 1\} = \{\alpha, 1+\alpha\}$, and both α and $1 + \alpha$ have degree 4 over $\mathbb{F}_2$. This contradiction shows $c = 1$, and hence $z = z'$.

Because the block containing 0 determines an additive coset partition, distinct subspaces zD give distinct round-two partitions. Thus the finite-field construction has no repeated round-two partition, whereas the SKC constructed here has a repeated round-two partition. The two SKCs cannot be equivalent.

5 Explicit seedings

For completeness, here is one direct seeding realization of the brackets. In each row, hyphens separate the four round-two blocks. Inside each block, the first two symbols and the last two symbols are the two first-round pairs. The first two hyphenated blocks form one semifinal half and the last two hyphenated blocks form the other.

```
T_1: 0123-89ab-4567-cdef
T_2: 0213-4657-8a9b-cedf
T_3: 03cf-12de-478b-569a
T_4: 04ae-268c-15bf-379d
T_5: 0527-8daf-1436-9cbe
T_6: 069f-178e-24bd-35ac
T_7: 07ad-349e-16bc-258f
T_8: 086e-3b5d-197f-2a4c
T_9: 095c-3a6f-184d-2b7e
T_a: 0a1b-4e5f-2839-6c7d
T_b: 0b6d-1a7c-294f-385e
T_c: 0c48-3f7b-1d59-2e6a
T_d: 0d58-1c49-2f7a-3e6b
T_e: 0e79-2c5b-1f68-3d4a
T_f: 0f4b-2d69-1e5a-3c78
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References

[1] R. Lambers, R. Pendavingh, and F. Spieksma. How to design a stable serial knockout competition. *Mathematics of Operations Research*, 50(2):1421–1432, 2025. doi:10.1287/moor.2022.0352.